

SAI KARTHIKEYA CHUKKA

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EDUCATION

PES University

B.Tech in CSE (AI & ML) — CGPA: 8.34/10 (as of July 2025)

Bangalore, India

Sep 2022 – Aug 2026

Sri Chaitanya Junior College

Class XII – MPC — 91.8%

Vijayawada, India

2020 – 2022

PROJECTS

Dockerized Real-Time Log Analytics Pipeline

- Built a single-node **real-time streaming pipeline** (Kafka → Spark Structured Streaming → JSON sinks → Streamlit) orchestrated with **Docker Compose**.
- Implemented **event-time analytics** with 1-minute windows, 2-minute watermark, and **durable host-mounted checkpoints** for error rate, latency, and status distribution metrics.
- Integrated Kafka via **spark-sql-kafka**; resolved file-sink issues with append+watermark; managed **topic initialization and offset resets** to recover from data loss.
- Reduced **cold start latency** by persisting Ivy/Maven cache and adding repository mirrors; exposed **Spark UI (4040)** and a minimal **Streamlit dashboard (8501)** for monitoring.

Event Management Web Application (DBMS Project) | Python, Flask, MySQL, Streamlit

2024

- Developed a **full-stack event management web app** with secure user authentication and event creation.
- Designed a **MySQL database** with triggers and stored procedures to enforce booking rules and ensure data integrity.
- Implemented a **Streamlit dashboard** for real-time visualization of events, bookings, and statistics.
- Created **Flask REST APIs** to support CRUD operations with comprehensive validation.

AI Question Paper Generation System | Python, NLP, RAG, Mistral, ChromaDB

2024–2025 (Ongoing)

- Built an AI-powered system to **auto-generate syllabus-aligned question papers** using a local Mistral model with RAG architecture.
- Integrated **textbook content into ChromaDB**, achieving **85% accuracy in CBSE-aligned question generation** without fine-tuning.
- Added **chunk diversity in retrieval** to reduce redundancy and improve semantic coverage of generated questions.
- Currently **fine-tuning with CBSE PYQs** and implementing a **feedback-driven refinement loop** to further enhance quality, accuracy, and diversity.

LungSimX: Pulmonary Drug Interaction Simulator | Three.js, JavaScript, CSS, HTML

2025

- Developed an interactive **3D lung simulation platform** to visualize drug effects on pulmonary tissue using Three.js.
- Implemented dynamic **drug selection and dosage controls** with real-time visualization of pharmacological impact and recovery.
- Created a **progress bar and timer** to represent effect duration, including color transitions and mesh-based spread/reversion animations.
- Designed a modern, responsive UI with animated backgrounds, iconography, and detailed drug information panels.
- Architected modular JavaScript code for maintainability and separation of UI and simulation logic.

CERTIFICATIONS

Databases and SQL for Data Science

Coursera – Authorized by IBM

Badge ID: 334e98b7-b523-493c-8ab9-042cd44dbff

Skills: SQL, Relational Databases, Python, Jupyter Notebook

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, JavaScript, HTML/CSS

ML/Data: TensorFlow, pandas, LangChain

DevOps: Docker, Kubernetes

Databases/Tools: MySQL, MongoDB